

Purpose of Credit models

We want to incorporate credit risk into our models to reflect real-world conditions.

Default time:

$$T_d = \inf \left(t \geq 0 \mid \int_0^t \lambda(s) ds \geq \phi \right)$$

$\lambda \rightarrow$ Stochastic Process, $\phi \sim \exp(1)$

Survival probability:

$$\mathbb{P}(T_d > t | \mathcal{G}_t) = e^{-\int_0^t \lambda(s) ds} \quad \forall t \geq 0$$

$\mathcal{G}_t \rightarrow$ All info in market except default info

After defining the framework we can arrive at some major summary Lemmas:

- ① Let X be a $\mathcal{G}_T$ -measurable random variable (and r is $\mathcal{G}$ -prog)

$$\mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s) ds} X 1_{\{T_d > T\}} | \mathcal{F}_t \right] = 1_{\{T_d > t\}} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T (r(s) + \lambda(s)) ds} X | \mathcal{F}_t \right]$$

SL1: Useful for pricing credit risk payments with fixed timing.

- ③ let $\{Y(t)\}_{t \geq 0}$ be a $\mathcal{G}$ - progressive stochastic process

$$\begin{aligned} & \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{T_d} r(s) ds} Y(T_d) 1_{\{t < T_d < T\}} \right] \\ &= 1_{\{T_d > t\}} \int_t^T \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^u (r(s) + \lambda(s)) ds} Y(u) \lambda(u) | \mathcal{F}_t \right] du \end{aligned}$$

SL2: Useful for pricing payoffs that happen at the default time T_d .

Types of recovery:

Zero-Recovery: Let X be a payment at time T . If the payer defaults nothing is recovered.

$$X_0(t, T) := \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r(s)ds} X | \mathcal{F}_t] \quad C_0(t, T) := \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r(s)ds} X 1_{\{T_d > T\}} | \mathcal{F}_t] = 1_{\{T_d > t\}} \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T (r(s) + \lambda(s))ds} X | \mathcal{F}_t]$$

Partial recovery at maturity: $R(t) \in (0, 1)$ is a stochastic process, that determines the amount recovered at maturity.

$$C(t, T) = \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r(s)ds} X (1_{\{T_d > T\}} + R(T) 1_{\{t < T_d \leq T\}}) | \mathcal{F}_t] = C_0(t, T) + \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r(s)ds} X R(T) 1_{\{t < T_d \leq T\}} | \mathcal{F}_t]$$

If we assume $R(T)$ is independent of r , λ and T_d :

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r(s)ds} X R(T) 1_{\{t < T_d \leq T\}} | \mathcal{F}_t] &= R(t, T) \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r(s)ds} X (1_{\{T_d > t\}} - 1_{\{T_d > T\}}) | \mathcal{F}_t] \\ &= R(t, T) (1_{\{T_d > t\}} X_0(t, T) - C_0(t, T)) \end{aligned}$$

Partial recovery at default: Assume instead recovery happens at default.

$$\begin{aligned} C_d(t, T) &= \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r(s)ds} X(T) 1_{\{T_d > T\}} + e^{-\int_t^{T_d} r(s)ds} X(T_d) R(T_d) 1_{\{t < T_d \leq T\}} | \mathcal{F}_t] \\ &= C_0(t, T) + 1_{\{T_d > t\}} \int_t^T \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^u (r(s) + \lambda(s))ds} X(u) R(u) \lambda(u) | \mathcal{F}_t \right] du \end{aligned}$$

Affine Processes with Credit Risk

Annoying term! Not affine!

$$r(t) = \bar{\rho}_0 + \bar{\rho}_1 \cdot Y(t)$$

$$\mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T (r(s) + \lambda(s)) ds} \lambda(T) | \mathcal{F}_t \right] = \mathbb{E}^{\mathbb{Q}} \left[\frac{\partial}{\partial x} e^{-\int_t^T (r(s) + \lambda(s)) du + x \lambda(T)} \Big|_{x=0} | \mathcal{F}_t \right]$$

$$\begin{aligned} \lambda(t) = \pi_0 + \pi_1 \cdot Y(t) &= \frac{\partial}{\partial x} \mathbb{E}^{\mathbb{Q}} [e^{-\int_t^T (r(s) + \lambda(s)) du + x \lambda(u)} | \mathcal{F}_t] \Big|_{x=0} = \frac{\partial}{\partial x} \left[e^{A(t, T, x) + B(t, T, x) \cdot Y(t)} \right] \Big|_{x=0} \\ &= (\partial_x A(t, T, 0) + \partial_x B(t, T, 0) \cdot Y(t)) e^{A(t, T, 0) + B(t, T, 0) \cdot Y(t)} \end{aligned}$$

Partial Recovery at Default: $X(t) := e^{h_0 + h_1 \cdot Y(t)}$ $R(t) := e^{j_0 + j_1 \cdot Y(t)}$

$$f(Y_t, t, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T (\rho_0 + \rho_1 \cdot Y(s)) ds + w_0 + w_1 \cdot Y(T)} | \mathcal{F}_t \right] = e^{A(t, T, x) + B(t, T, x) \cdot Y(t)}$$

$$\rho_0 = \bar{\rho}_0 + \pi_0, \quad \rho_1 = \bar{\rho}_1 + \pi_1, \quad w_0 = h_0 + j_0 + x \pi_0, \quad w_1 = x \pi_1 + h_1 + j_1$$

$$\begin{aligned} C_D(t, T) &= 1_{\{T_d > t\}} \int_t^T \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^u (r(s) + \lambda(s)) ds} X(u) R(u) \lambda(u) | \mathcal{F}_t \right] du \\ &= 1_{\{T_d > t\}} \int_t^T \frac{\partial}{\partial x} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^u (r(s) + \lambda(s)) ds + x \lambda(u)} X(u) R(u) | \mathcal{F}_t \right] \Big|_{x=0} du \\ &= 1_{\{T_d > t\}} \int_t^T (\partial_x A(t, u, 0) + \partial_x B(t, u, 0) \cdot Y(t)) e^{A(t, u, 0) + B(t, u, 0) \cdot Y(t)} du \end{aligned}$$

Credit Default Swaps

Insurance against a default event, where coupon payments are made regularly in return for a payment that pays out Loss if the underlying defaults.

Coupon payments up until Default: $I_1(t) = \mathbb{E}^{\mathbb{Q}}[\sum_{i=1}^n e^{-\int_t^{T_i} r(s)ds} \delta_i 1_{\{T_d > T_i\}} | \mathcal{F}_t], \quad \delta_i = T_i - T_{i-1}$

The remaining Coupon payment if
Default happens during a period:

$$I_2(t) = \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^{T_d} r(s)ds} (T_d - T_{\eta(T_d)-1}) 1_{\{T_0 < T_d \leq T\}} | \mathcal{F}_t]$$

index of the T after default occurred



$$V_{prem}(t) = C(I_1(t) + I_2(t)) \quad V_{prot} = \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^{T_d} r(s)ds} (1 - R(T_d)) 1_{\{T_0 < T_d \leq T\}} | \mathcal{F}_t]$$

The value of C that makes them equal is the CDS-spread: $CDS(t, T) := C = \frac{V_{prot}(t)}{I_1(t) + I_2(t)}$

Constant λ (unrealistic assumption) implies high CDS-spread $\rightarrow$ High probability of default. Take with grain of salt.

Multicurve models

xIBOR rate carries significant credit risk as shown in GFC.

$$IRS(t, T_n) = \frac{\sum_{i=1}^n \delta_i \mathbb{E}_t^Q \left[e^{-\int_t^{T_i} r_c(u) du} L(T_{i-1}, T_i) \right]}{\sum_{i=1}^n \delta_i P^c(t, T_i)}$$

Filipovic and Trolle want a model that can decompose $\underbrace{x\text{IBOR}}_{\leq \text{IRS}} = f(\underbrace{\text{Interest Rate Risk}}_{\leq \text{OIS}}, \underbrace{\text{Credit Risk}}_{\leq \text{CDS}}, \underbrace{\text{Liquidity Risk}}_{\leq \text{Residual}})$

They propose the following 6-dim affine model:

$$dr(t) = \kappa_r(\gamma(t) - r(t))dt + \sigma_r dW_r(t),$$

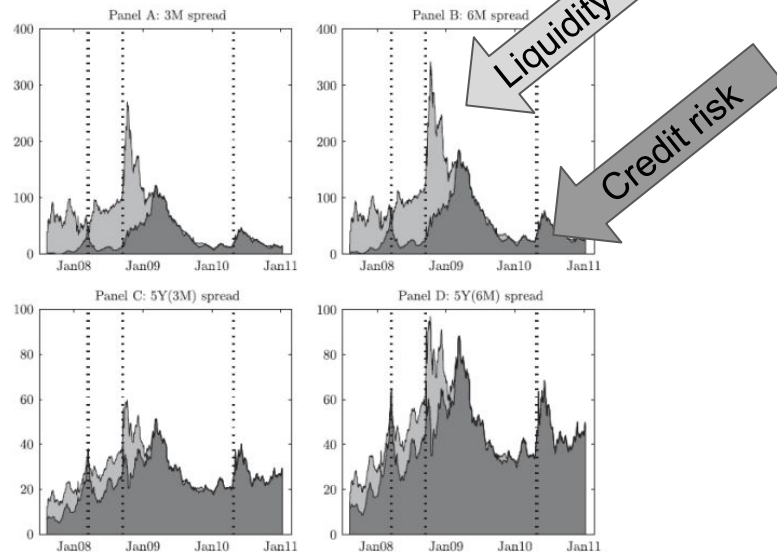
$$d\gamma(t) = \kappa_\gamma(\theta_\gamma - \gamma(t))dt + \sigma_\gamma(\rho dW_r(t) + \sqrt{1 - \rho^2} dW_\gamma(t))$$

$$d\lambda(t_0, t) = \kappa_\lambda(\Lambda - \lambda(t_0, t))dt + dJ_\lambda(t_0, t)$$

$$d\nu(t) = \kappa_\nu(\mu(t) - \nu(t))dt + \sigma_\nu \sqrt{\nu(t)} dW_\nu(t),$$

$$d\mu(t) = \kappa_\mu(\theta_\mu - \mu(t))dt + \sigma_\mu \sqrt{\mu(t)} dW_\mu(t),$$

$$d\xi(t) = \kappa_\xi(\theta_\xi - \xi(t))dt + \sigma_\xi \sqrt{\nu(t)} dW_\xi(t).$$



Banks were afraid to lend short-term because they were scared of not being liquid enough, which drove LIBOR rates up!